Mackerel Road: Learning Neural Game Mechanics Without Transition Targets

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Abstract

Can we learn a game's mechanics from desired behavior rather than examples of correct transitions? We study *Mackerel Road*, a stochastic fish-delivery race whose local neural network defines travel costs and events. Differentiable planning trains this rule from objectives for viable deliveries, role balance, competence response, and weather, without a transition teacher. A matched four-stage ablation uses three optimizer seeds and 100 held-out maps per seed. The full model retains viable deliveries and stochastic events, but its sampled competence curve is steeper than the design target. Fitting a differentiable objective is not the same as achieving its sampled-play counterpart or human enjoyment.

1. Introduction

Search-based game design optimizes content or rules using simulated play (Togelius et al., 2011; Hom & Marks, 2007). Restricted players reveal balance problems (Jaffe et al., 2012); controlled mistakes probe competence response (Graham et al., 2023). The author's ongoing, unpublished *Computational Game Quality* (CGQ) work treats this response curve as an evaluation object: how does a game turn degraded decisions into worse outcomes? Here we use such a curve as a *design objective* for a neural mechanic.

The network is not predicting a pre-existing game's rules: it defines the local transition used in play. The contribution is a course-scale test of the chain *neural rule* $\rightarrow$ *differentiable player* $\rightarrow$ *game-level loss* $\rightarrow$ *held-out sampled play*. A constrained game grammar prevents arbitrary outcome assignment; a loss ablation tests which desired properties survive deployment. Code, selected weights, and measured results: `https://github.com/cem-sirin/dlai-project`.

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2. Method

Game. A layered route map gives light and strong porters 7–10 sequential road choices. Roads differ in distance, difficulty, exposure, and terrain. A visible cloud moves after each round and rains on incoming roads. Porters choose independently on the same fixed map; neither changes the other's transitions. Travel consumes stamina and fish freshness; exhaustion fails the delivery, and higher terminal freshness wins. This small setting permits full-route planning while retaining route trade-offs and stochastic delays (Appendix A).

Local transition network. An 8,252-parameter SiLU MLP system receives 12 local features: role, own stamina and freshness, cargo, road attributes, and rain. It excludes map topology, position, opponent, and distance to the goal, preventing direct winner control. Bounded heads predict time T, stamina loss S, spoilage rate R, event probabilities, and delays δ_k. Difficulty gates favor strong on hard roads and light on easy roads. Role-blind events are no event, landslide, fallen tree, or cart blockage, with terrain masks. The sampled transition is

$$\begin{aligned} t' &= t + T_\theta(x) + \delta_k, \\ a' &= [a - S_\theta(x)]_+, \\ f' &= f \exp[-R_\theta(x)(T_\theta(x) + \delta_k)]. \end{aligned} \tag{1}$$

The grammar and bounds are hand-designed; the state/road dependence within them is learned, not copied from transition labels.

Differentiable player. The full-map player expands road and cloud branches, continuing from the mean immediate event state. Softmax road choices and smooth terminal survival make expected delivery score differentiable. To probe competence, mix a fixed uncorrupted reference policy with uniform legal play:

$$\pi_\epsilon = (1 - \epsilon)\pi_0 + \epsilon U, \qquad \epsilon \in [0, 1]. \tag{2}$$

Only the focal player is degraded; both role assignments are averaged. Hard evaluation uses argmax choices and samples corruption, clouds, and events. The reference is full-horizon but not an exact stochastic-game solver (Appendix B).

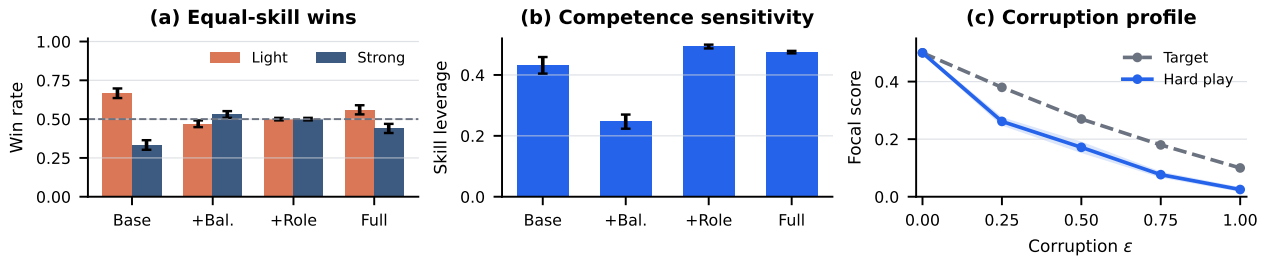


Figure 1. Matched held-out hard play on the same 100 maps per model seed. Base includes viability, luck, and regularization; subsequent stages add balance, role specialization, then skill and weather jointly. (a) Role win rates; (b) skill leverage; (c) full-model focal score versus the design target. Bars and shading are population SD across three model seeds, not confidence intervals.

3. Learning objective

For batches of random maps, PyTorch autograd (Paszke et al., 2019) minimizes

$$\min_{\theta} \mathbb{E}_{\mathcal{B}}[\mathcal{L}_{\text{viable}} + \mathcal{L}_{\text{balance}} + \mathcal{L}_{\text{role}} + \mathcal{L}_{\text{skill}} + \mathcal{L}_{\text{weather}} + \mathcal{L}_{\text{luck}} + \mathcal{L}_{\text{reg}}]. \tag{3}$$

Why these losses? Viability uses squared hinges to reject both failed deliveries and cost-free perfect delivery. Balance penalizes the squared mean role-score gap; specialization targets advantages in both directions across maps, avoiding fairness by identical outcomes. Map statistics are pooled before these nonlinear penalties. Skill minimizes scaled squared error between a sigmoid comparison of expected scores and targets $(.50, .38, .27, .18, .10)$ at $\epsilon = (0, .25, .5, .75, 1)$. This encodes rewarding good choices while preserving some chance for random play, not an empirically optimal enjoyment curve.

Weather targets a .025 cloud-induced score cost so the visible cloud matters. Luck bounds event rate and delay variance to prevent deterministic shortcuts. Regularization penalizes counterintuitive local responses and large weights. Thus there are *game-level targets*, but no *transition targets*. Exact formulas, scales, and limitations are in Appendix C.

4. Experiments and results

We use 64 training, 16 validation, and 100 disjoint test maps, shared across three optimizer seeds. Each variant starts from initialization. AdamW uses $\eta = .003$, weight decay 10^{-5}, four maps per update, and at most 120 updates on an RTX 5090. Only validation loss selects checkpoints. Equal competence means both players have $\epsilon = .2$; the profile degrades one against the uncorrupted reference. The focal score $g(\epsilon)$ is the win rate plus half the draw rate, not terminal freshness; skill leverage is $g(0) - g(1)$.

Figure 1 and Appendix D report the matched ablation. Base favors light (66.7 ± 3.1 percent wins). Balance reduces that bias but lowers leverage from 0.432 ± 0.027 to 0.247 ± 0.023; adding role specialization yields equal-competence parity (50.0 ± 0.8 percent light wins) and 0.493 ± 0.006 leverage. Full does not dominate every metric: jointly targeting skill and weather trades away some of that parity.

Full-model light wins are 56.0 ± 2.9 percent, final freshness is 0.514 ± 0.039, and leverage is 0.475 ± 0.004, with no observed delivery failures in equal-competence races. Validation cloud cost is 0.0283 ± 0.0011 (target .025). The random focal player's score is 0.025, compared with the .10 target. Sampled events occur on 17.2 ± 0.8 percent of equal-competence traversals. These measurements test the deployed mechanic, rather than merely reporting a decrease in its differentiable training loss.

5. Discussion

The result supports optimizing a constrained neural mechanic through aggregate play objectives without transition supervision. It does not establish human enjoyment, globally optimal players, or perfect attainment of every target. Soft training versus sampled play and finite semantic probes remain important limitations. Skill and weather are added jointly, so the ablation cannot isolate their individual effects. A stronger competence gradient also need not mean greater tolerance of mistakes. Appendix E exposes per-seed variation, residual losses, and the surrogate–hard-play gap rather than hiding them behind a single aggregate score.

Statement on the use of AI

Hermes Agent was used as an AI-assisted tool to help write scripts, generate plots, and draft and edit the report. The project ideas and methodology are the author's. The author directed the work, reviewed all code, and remains responsible for the results and the submitted material.

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A. Game specification and architectural choices

Why this game? The fish-delivery race is small enough to expand complete routes, yet contains sequential consequences: an early road choice changes the stamina and freshness available later. Branching and rejoining create alternatives without requiring an open-ended action space. Two porter roles make balance measurable; a visible moving cloud makes adaptation possible; sampled delays preserve uncertainty. The map is regenerated between races, so the learned local rule must operate on unseen combinations of roads rather than memorize a single board. The two porters do not block, attack, or otherwise change one another's transitions. Competition is a comparison of deliveries under shared weather, not a model of strategic opponent interaction.

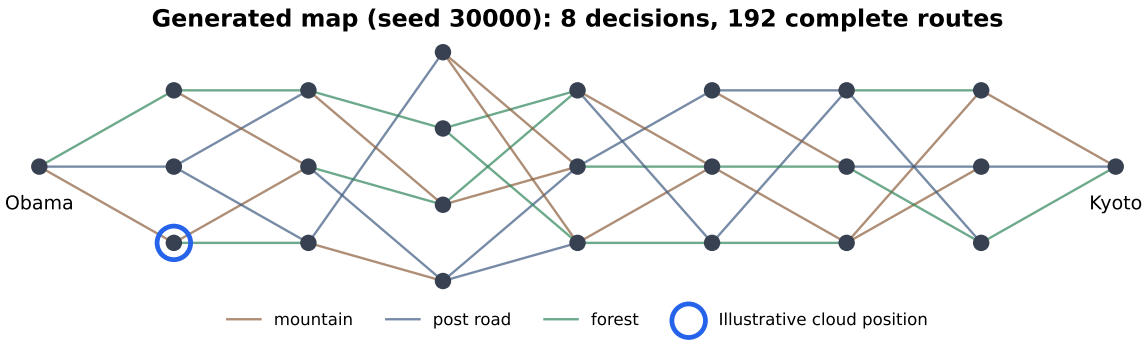


Figure 2. An actual generated test map; roads go from left to right, one layer at a time. Colors denote terrain, not learned cost. The ring illustrates one legal initial cloud location, not a measured player trajectory. The graph is shared by both porters and remains fixed during a race.

Map and weather distribution. The horizon is uniform on $\{7, 8, 9, 10\}$. The first and last internal layers have three nodes; the others have three to five. Every destination node receives an incoming edge; additional links give each source at least two distinct outgoing roads where possible, with a single common terminal node. Terrain labels are approximately balanced within each layer transition. Conditional attributes are sampled independently from the intervals below and rounded to three decimals. The cloud starts uniformly in the first internal layer, follows a uniformly sampled outgoing road after each round, and disappears when it reaches the terminal node. A road is rainy exactly when its destination is the current cloud node. Exposure is an additional continuous input, not the cloud's transition probability.

Terrain	Distance	Difficulty	Exposure	Allowed delay event
Mountain	$U(1.8, 3.8)$	$U(.65, .95)$	$U(.55, .95)$	Landslide
Forest	$U(2.4, 4.5)$	$U(.30, .70)$	$U(.15, .55)$	Fallen tree
Post road	$U(3.2, 5.5)$	$U(.05, .35)$	$U(.25, .65)$	Cart blockage

Information boundary and architecture. The local feature vector is $x = (r_L, r_S, a, f, c, d/5.5, h, e, m_M, m_F, m_P, w)$: role bits, normalized stamina, freshness, cargo, normalized distance, difficulty, exposure, terrain bits, and rain. Cargo is fixed at .75. All heads have two hidden layers with SiLU activations and a linear output layer. Their dimensions are:

Head	Inputs	Layer widths	Output meaning
Base	$a, f, c, d/5.5, h, e, m_M, m_F, m_P, w$	$10 \to 64 \to 64 \to 3$	Time, stamina loss, spoilage rate
Role	h	$1 \to 16 \to 16 \to 2$	Speed and stamina advantages
Event	$d/5.5, h, e, m_M, m_F, m_P, w$	$7 \to 32 \to 32 \to 4$	Event logits
Delay	Same as event head	$7 \to 32 \to 32 \to 3$	Real-event delays

The model has 8,252 trainable parameters. A local MLP is sufficient for these fixed-size road/state features; a graph network would unnecessarily expose the mechanic to global position and topology. The *planner*, in contrast, knows the graph and cloud state. The base and event heads do not see role; role-dependent behavior enters through explicit difficulty

gates. This prevents direct role-dependent event probabilities, but roles can still encounter different event frequencies by choosing different roads.

B. Transition equations and differentiable planning

A constrained neural rule, not unconstrained rule discovery. Writing σ for the logistic sigmoid and (b_T, b_S, b_R) for the base-head outputs, define

$$T_0 = .20 + 5.80\sigma(b_T), \qquad S_0 = .16\sigma(b_S), \qquad R = .060\sigma(b_R), \tag{4}$$

$$A_T = .55\sigma(u_T(h)), \qquad A_S = .55\sigma(u_S(h)), \tag{5}$$

$$T = T_0 \exp[-A_T\{r_S h + r_L(1-h)\}], \qquad S = S_0 \exp[-A_S r_S h]. \tag{6}$$

Thus light is locally faster on easy roads and strong on hard roads, at otherwise identical inputs; only strong receives an explicit stamina reduction. These semantics are architectural constraints, not empirical discoveries. Event probabilities are a softmax with incompatible terrain events masked to zero; no event is always allowed. The delay head gives $\delta_0 = 0$ and $\delta_k = 3\sigma(z_k)$ for each real event. On branch k,

$$(t', a', f') = \big(t + T + \delta_k,\ \mathrm{clip}(a - S, 0, 1),\ f \exp[-R(T + \delta_k)]\big). \tag{7}$$

Delay affects time and freshness, not stamina directly. Exhaustion is absorbing in sampled play, with terminal freshness set to zero. The winner has higher final freshness; a difference within 10^{-9} is a draw. The evaluator stores physical stamina initialized to 15 and 17, but normalizes before every neural call and rescales afterward; the normalized dynamics start at one for both roles.

Initialization and inductive bias. Output-layer weights start at zero. Biases initialize $T_0 = 2.4$, $S_0 = .065$, $R = .018$, $A_T = A_S = .20$, compatible-event probability .20, and event delay .80. Hidden weights use PyTorch initialization. This starts from a viable, stochastic rule rather than arbitrary destructive transitions. The DAG, action space, freshness update, output bounds, terrain masks, and role gates are all hand-designed. What optimization learns is the local dependence of costs and events on state, road, and weather within this grammar. There are no observed next-state labels or teacher-engine regression targets.

Planner and gradient path. Let $\bar{s}'_\theta(s, a)$ contain the immediate event-averaged next stamina, freshness, and elapsed time. In particular, $\bar{f}' = \sum_k p_k f \exp[-R(T + \delta_k)]$; the exponential is taken before averaging. The planner explicitly expands all road and cloud branches but continues from this *mean event state*, not every sampled event history. At a terminal state it uses the smooth score

$$v(a, f) = f\, \sigma((a - .025)/.01)\, \sigma((f - .025)/.01). \tag{8}$$

Backward recursion defines uncorrupted action values Q_0, reference choices $\pi_0(a \mid s) = \mathrm{softmax}(35Q_0(s, \cdot))_a$, and

$$Q_\epsilon(s, a) = \sum_{z'} P(z' \mid z) V_\epsilon(\bar{s}'_\theta(s, a), z'), \tag{9}$$

$$V_\epsilon(s) = \sum_a \big[(1 - \epsilon)\pi_0(a \mid s) + \epsilon U(a \mid s)\big] Q_\epsilon(s, a). \tag{10}$$

The same π_0 is used at every corruption level; it is not reoptimized to anticipate future mistakes. Gradients pass through neural transitions, event probabilities, expected states, and soft road choices into the game-level loss. This is differentiable optimization of a mechanic, not supervised transition prediction or reinforcement learning of a separate player network.

What hard evaluation changes. Hard play recomputes an uncorrupted, full-horizon argmax choice from the actual current state, then replaces it with a uniformly sampled legal road with probability ϵ. A replacement can coincide with the reference action: ϵ is a replacement probability, not the fraction of objectively wrong moves. Clouds and events are sampled, and failures stop a porter. The reference planner still uses mean future event states; "uncorrupted" does not mean an exact solution of the full stochastic game or an optimal win-probability policy. The train/evaluation differences include soft versus hard decisions, smooth versus absorbing survival, a sigmoid score-comparison surrogate versus sampled wins, and expected versus realized future states.

C. Loss design and numerical specification

These objectives express a chosen game design. They are not estimates of a universal definition of quality, nor labels inferred from human enjoyment. Squared errors give gradients proportional to target mismatch; squared hinge penalties accept a *range* of viable or stochastic behavior without forcing all games to one value. Scales below normalize heterogeneous quantities and set their relative influence. They are fixed design hyperparameters, not the result of a hyperparameter search.

Let $v_{r,M}(\epsilon)$, $f_{r,M}(\epsilon)$, and $a_{r,M}(\epsilon)$ be planner outputs, averaged over initial cloud positions. A bar denotes the mean over sampled maps, and over roles where appropriate. Let $[x]_+ = \max(x, 0)$, $d_M = v_{S,M}(0) - v_{L,M}(0)$, and $s(x) = .01 \log(1 + \exp(x/.01))$.

Viability: avoid both free delivery and unavoidable failure. The acceptable terminal score interval is $[.22, .72]$, with at least .10 normalized stamina. Using uncorrupted planner outputs,

$$\mathcal{L}_{\text{viable}} = \overline{\left(\frac{[v - .72]_+}{.10}\right)^2 + \left(\frac{[.22 - v]_+}{.10}\right)^2 + \left(\frac{[.10 - a]_+}{.10}\right)^2} \tag{11}$$

$$+ .10\,\overline{\sigma((.025 - f)/.01) + \sigma((.025 - a)/.01)}. \tag{12}$$

Simply maximizing freshness would permit a trivial mechanic in which every route delivers perfectly. The upper viability bound avoids that solution; minimum score and stamina discourage the opposite collapse.

Balance and specialization: fairness without identical roles.

$$\mathcal{L}_{\text{balance}} = (\bar{d}/.025)^2, \tag{13}$$

$$\mathcal{L}_{\text{role}} = \left(\frac{\overline{s(d)} - .020}{.015}\right)^2 + \left(\frac{\overline{s(-d)} - .020}{.015}\right)^2. \tag{14}$$

Balance concerns the mean *score gap*, not directly the win rate. The role term asks for advantages in both directions across maps, rather than equal outcomes on every map. Softplus supplies smooth positive-part estimates, but $s(0) = .01 \log 2 > 0$: a positive diagnostic alone does not prove specialization. We therefore also report signed-gap fractions and mean absolute gaps. Crucially, map statistics are pooled *before* squaring; averaging squared singleton losses would impose a different objective.

Competence response: reward decisions without demanding determinism. For $\epsilon_j \in \{0, .25, .50, .75, 1\}$, the differentiable focal score is

$$\hat{g}_j = \frac{1}{2}\overline{\sigma\left(\frac{v_L(\epsilon_j) - v_S(0)}{.025}\right) + \sigma\left(\frac{v_S(\epsilon_j) - v_L(0)}{.025}\right)}, \qquad \mathcal{L}_{\text{skill}} = \frac{1}{5}\sum_j \left(\frac{\hat{g}_j - y_j}{.10}\right)^2, \tag{15}$$

with $y = (.50, .38, .27, .18, .10)$. This deliberately declining target encodes a preference that better choices matter, while a random focal player retains some chance against the reference. Its particular shape is not theoretically optimal. The sigmoid is a differentiable comparison of expected scores, not the expected win indicator; matching it need not exactly match the hard-play profile.

Weather: a perceptible but bounded challenge. Let $\Delta_w = \overline{v_r^{\text{clear}}(0) - v_r^{\text{cloud}}(0)}$, where the clear counterfactual disables clouds but retains event randomness. Then $\mathcal{L}_{\text{weather}} = ((\Delta_w - .025)/.015)^2$. This prevents the network from simply ignoring the visible cloud. The target is an absolute score reduction, not a relative percentage change.

Luck and semantic regularization: close alternative shortcuts. On 96 fixed, deterministic local feature probes, let $\rho = \overline{1 - p_0}$ and $V = \overline{\sum_k p_k \delta_k^2 - (\sum_k p_k \delta_k)^2}$. We use

$$\mathcal{L}_{\text{luck}} = \left(\frac{[.12 - \rho]_+}{.10}\right)^2 + \left(\frac{[\rho - .35]_+}{.10}\right)^2 + \left(\frac{[.04 - V]_+}{.10}\right)^2 + \left(\frac{[V - .80]_+}{.50}\right)^2. \tag{16}$$

Both frequency and variance matter: frequent zero-delay events would otherwise satisfy an event-rate constraint without consequential randomness. These are probe-distribution quantities, not policy-weighted traversal frequencies.

Counterfactual probes increase normalized distance or difficulty by .12, switch rain off/on, or compare roles at difficulty .85 and .15. One-sided ReLU penalties discourage reduced travel/stamina cost on longer or harder roads, reduced expected travel time or spoilage rate under rain, and reversed role advantages. The sum is weighted by .25. In addition, 10^{-5} times the mean of parameter-tensor mean squares is added; AdamW separately uses weight decay 10^{-5}. Probe constraints are soft and finite, not global monotonicity proofs.

D. Experimental protocol, complete results, and interpretation

A bounded experiment. All four nested variants use optimizer seeds $0, 1, 2$, the same 64 training maps (seeds 0–63), 16 validation maps (10000–10015), and 100 test maps (30000–30099). Training uses AdamW at .003, batches of four maps sampled without replacement within each update, gradient-norm clipping at 5, at most 120 updates, validation every ten updates, and patience six validation checks. Validation evaluates the pooled objective on the whole validation set. Each seed's checkpoint minimizes its own variant's validation loss; no test metric selects checkpoints. All variants are trained from initialization, not continued from the preceding variant. Different objectives have different numerical scales, so their raw total losses are not directly comparable.

All reported figures and tables use this matched experiment. The test seed range is disjoint from the 20000–20099 range used in preliminary experiments; those earlier results are not mixed into the reported measurements. No further hyperparameter search was performed for this submission.

Execution was recovered at completed-seed boundaries after a wall-clock limit and a final-seed GPU-memory failure. The last seed uses per-plan activation checkpointing, verified against ordinary backpropagation for loss and gradient agreement, and runs in tmux. This changes the memory/computation trade-off, not the objective or batch size; incomplete attempts are excluded from per-run times.

Table 1. Matched held-out ablation. Values are means ± population SD over three trained seeds, each evaluated on the same 100 maps. All entries are fractions or normalized scores, not percentages. Base includes viability, luck, and regularization; later stages add balance, role specialization, and jointly skill plus weather.

Variant	Light wins	Freshness	Event rate	Leverage	Tolerance
Base	0.667 ± 0.031	0.523 ± 0.054	0.258 ± 0.011	0.432 ± 0.027	0.390 ± 0.016
+Balance	0.470 ± 0.022	0.644 ± 0.023	0.233 ± 0.014	0.247 ± 0.023	0.478 ± 0.024
+Role	0.500 ± 0.008	0.434 ± 0.016	0.182 ± 0.028	0.493 ± 0.006	0.293 ± 0.026
Full	0.560 ± 0.029	0.514 ± 0.039	0.172 ± 0.008	0.475 ± 0.004	0.354 ± 0.011

What the estimators measure. Equal-competence races corrupt both porters at $\epsilon = .2$; reference races corrupt neither. For every corruption-profile point, one focal role is degraded and the other remains uncorrupted; both focal assignments are averaged. Score is 1 for a focal win, $1/2$ for a draw, and 0 for a loss. At $\epsilon = 0$, the two assignments reproduce the same match with complementary focal scores; $g(0) = .5$ follows by construction and is *not* evidence of role balance. We instead measure balance through the separate light/strong win rates.

The evaluator uses one seeded realization per map and competence condition, with cloud, event, and decision random streams shared across model seeds and variants. Profile streams are reused across corruption levels. Reusing seeds improves comparability but does not turn the model seeds into independent sets of maps. Error bars are descriptive across-model SD, not confidence intervals or significance tests. Mean freshness, failure rates, and event frequencies refer specifically to the equal-competence races; zero failures there is not a guarantee about all possible policies or trajectories.

Leverage is not tolerance. The CGQ-inspired endpoint summary is $L = g(0) - g(1)$. When $L > 0$, normalized retention is $r(\epsilon) = (g(\epsilon) - g(1))/L$ and tolerance is $T = \int_0^1 r(\epsilon)d\epsilon$, estimated by the trapezoidal rule on the five reported points. A large leverage means this execution-competence axis affects outcomes; it does not imply forgiving gameplay. A low tolerance indicates that much of the advantage is lost early under this particular intervention. These are policy- and context-dependent signatures, not intrinsic human skill ratings.

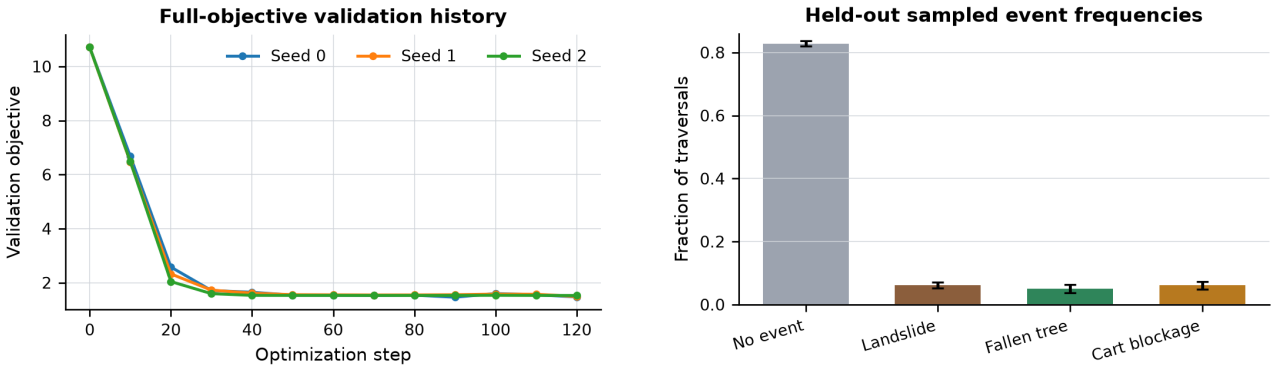


Figure 3. Left: full-objective validation histories used for checkpoint selection. Right: sampled event frequencies in held-out equal-competence races; error bars are population SD across model seeds. Terrain frequencies and route choices affect these counts, so they need not equal the local probe rate.

E. Per-seed evidence and remaining limitations

Table 2. Every executed final run. Seconds include training, validation, and hard evaluation, not just optimizer time. Failure fractions are measured in equal-competence races. The distributed full-model checkpoint is seed 0, selected by validation loss rather than test performance.

Variant	Seed	Best step	Validation loss	Light wins	Failures	Seconds
Base	0	60	0.0000	0.650	0.000	112.1
Base	1	30	0.0000	0.710	0.000	102.5
Base	2	30	0.0000	0.640	0.000	103.1
+Balance	0	70	0.0000	0.480	0.000	110.4
+Balance	1	120	0.0000	0.490	0.000	111.4
+Balance	2	110	0.0000	0.440	0.000	110.3
+Role	0	90	0.7253	0.490	0.000	110.6
+Role	1	120	0.8318	0.500	0.000	110.9
+Role	2	120	0.6844	0.510	0.000	110.4
Full	0	90	1.4610	0.600	0.000	135.8
Full	1	120	1.4877	0.550	0.000	133.4
Full	2	120	1.5246	0.530	0.000	160.9

Interpreting the residuals. Figure 4 shows that the validation surrogate closely follows the chosen competence target while sampled play penalizes corruption more strongly. The full-model tolerance is 0.354 ± 0.011: high leverage should not be mistaken for gentle degradation. Weather cost is near its target, but the role-specialization term remains a major residual. Its softplus advantages remain below .020, and the small absolute score gaps do not establish robust map-dependent specialization. Difficulty penalties also remain positive: soft local regularization has not guaranteed monotone behavior. These are observable limits of this short optimization, not properties the report assumes away.

What this establishes, and what it does not. The experiment tests whether a constrained local neural rule can be optimized through differentiable game-level objectives and then used in sampled play on held-out maps. It does not show that every loss target is achieved, that specialization is robust to a different map distribution, or that the resulting game is enjoyable for humans. Only three optimizer seeds, a short training budget, and one stochastic realization per map-condition are used. The nested ablation measures incremental changes in this order; skill and weather enter together, so their individual causal effects are not identified. The absence of a transition teacher does not remove human design choices from the grammar, initialization, or loss. Useful next steps are matched stochastic estimators, more weather/event repetitions, a factorial loss ablation, held-out role and cloud counterfactuals, and human playtesting.

Reproducibility. The repository contains the selected state dictionary in `results/selected_model.json`, all final protocol/history/metric records in `results/neural-evaluated.json`, and the figure/table builder

Table 3. Full-model diagnostics at the selected checkpoints. Role and weather quantities are validation-surrogate measurements on 16 maps; the difficulty penalty is computed on the fixed semantic probe set. They are not additional held-out hard-play measurements.

Quantity	Seed 0	Seed 1	Seed 2
Mean role gap (strong minus light)	-0.0031	0.0014	-0.0014
Mean absolute role gap	0.0065	0.0059	0.0027
Strong-favored map fraction	0.2500	0.5000	0.2500
Light-favored map fraction	0.7500	0.5000	0.7500
Cloud cost	0.0288	0.0292	0.0267
Difficulty penalty	0.0552	0.0607	0.0557

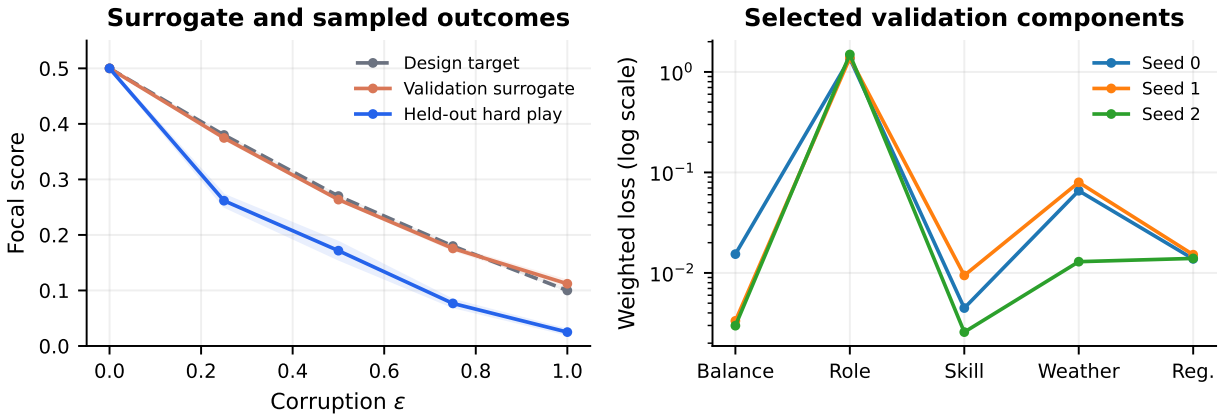


Figure 4. Left: the design target, selected validation-surrogate profiles, and held-out hard profiles; shading is across-seed population SD. Different map sets and multiple estimator differences contribute to the gap, so this is not an isolated measurement of event-averaging bias. Right: remaining weighted loss components at the selected full checkpoints. Components are nonnegative; viability and luck can be inactive inside their acceptable ranges.

`mackerel_road.experiments.submission`. The reporting path uses stored measurements and requires no GPU or PyTorch. The full training command and verification instructions are in `README.md` and `report/README.md`. Raw run payloads retain all seed checkpoints locally; only the selected checkpoint is included in the submission repository.